10:13 4G 63

Not yet answered

Marked out of 1.00

Flag question

Time left 0:07:21

For single-phase full converter drive of sep.excited dc motor, the reversal of the armature or field allows operation in the

- a. second and third quadrants
- b. first and third quadrant
- c. third and fourth quadrant
- d. first and second quadrant

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◄ Quiz 1

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Time left

Question 5

Not yet answered

Marked out of 1.00

Flag question

Varying stator voltage and stator frequency together, if $T_L=k\,\omega^2$, Vn=240V, fn=50Hz, the voltage at f=30Hz should be

a. 240V

b. 86.4V

c. 144V

d. 186 V

10:15 4G 60

Question 8

Time left 0:04:49

Not yet answered

Marked out of 1.00

Flag question

Motor duty is considered continuous duty (S1), if the motor operates with the following conditions

a. temperature rise during operation under load does not reach steady-state value

b. temperature rise during operation under load reaches steady-state value

c. starting time is not neglected

d. starting losses are not neglected

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◄ Quiz 1

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Question **3**

Not yet answered

Marked out of 1.00

Flag question

Mechanical characteristics of drive system is

- a. ω=f(N)
- b. ω=f(I)
- c. ω=f(t)
- d. ω=f(T)

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10:09 4G 64

Question 3

Not yet answered

Marked out of 1.00

Flag question

Time left 0:11:38

With a single –phase full-wave converter in the armature circuit, the average armature voltage is

a. $V_a = \frac{2V_m}{\pi} cos\alpha$

b. $V_a = \frac{V_m}{2\pi} cos\alpha$

c. $\frac{V_m}{\pi}(cos\alpha + 1)$

d. $V_a = \frac{2V_m}{\pi}(cos\alpha + 1)$

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10:04 78%
s2.ebalqa.courses/nfe 24

Time left 0:11:16

Question 6

Not yet answered

Marked out of 1.00

Flag question

In four quadrant operation of a motor driving a hoist load(elevator), if the cage is loaded and the motor speed is downward, then the motor is working in

a. the third quadrant

b. the fourth quadrant

c. the first quadrant

d. the second quadrant

Clear my choice

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10:09 4G 63

Time left 0:10:49

Question **4**

Not yet answered

Marked out of 1.00

Flag question

The temperature rise is related to ambient temperature which value is equal to

a. 20 °C

b. 24°C

c. 30 °C

d. 40 °C

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10:02 78%
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Question 3

Time left 0:12:47

Not yet answered

Marked out of 1.00

Flag question

Torque equation for rotational motion when inertia is variable

a. $F_m - F_f = \frac{d(Mv)}{dt}$

b. $T - T_L = J\frac{d\omega_m}{dt}$

c. $T - T_L = J\frac{d^2\theta}{dt^2}$

d. $T - T_L = J\frac{d\omega_m}{dt} + \omega_m\frac{dJ}{dt}$

Clear my choice

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← week 14

Jump to...

10:02 79%
s2.ebalqa.courses/nfe 24

Time left 0:13:27

Question 2

Not yet answered

Marked out of 1.00

Flag question

Torque equation for rotational motion when inertia is constant

a. $T - T_L = J\frac{d\theta}{dt}$

b. $T - T_L = J\frac{d\omega_m}{dt} + \omega_m\frac{dJ}{dt}$

c. $F_m - F_f = \frac{d(Mv)}{dt}$

d. $T - T_L = J\frac{d^2\theta}{dt^2}$

Clear my choice

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← week 14

Time left 0:01:56

Varying stator voltage and stator frequency together, if $T_L = k_1/\omega$, Vn=240V, fn=50Hz, the voltage at f=30Hz should be

- a. 240V
- b. 186V
- c. 86.4V
- d. 144V

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10:08 4G 66

Time left 0:12:19

Question **2**

Not yet answered

Marked out of 1.00

Flag question

The average output of a single-phase semiconverter connected to 120V, 50Hz,supply and firing angle of π/2 is

- a. 54.02V
- b. 27V
- c. 0V
- d. 120V

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◄ Quiz 1

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10:16 4G 59

Marked out of 1.00

Flag question

Time left 0:04:16

The temperature limit of insulating materials class c is

- a. 150 °C
- b. > 180 °C
- c. 180 °C
- d. 130 °C

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Finish attempt ...

◄ Quiz 1

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Time left 0:12:35

Question **2**

Not yet answered

Marked out of 1.00

Flag question

The active load torque industrial machinery is

- a. compressor
- b. conveyor
- c. hoisting mechanism
- d. turning machine

Clear my choice

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Time left 0:09:49

Question 9

Not yet answered

Marked out of 1.00

Flag question

The hardness of series field dc motors torque-speed characteristic is

- a. negative
- b. zero
- c. positive
- d. infinity

Clear my choice

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2024/2023-2-ELE2461-ELECTRICAL DRIVES-2

Time left 0:03:06

Quiz navigation

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Finish attempt ...

Question 8

Not yet answered

Marked out of 1.00

Flag question

Varying stator voltage and stator frequency together, if TL=Tn, Vn=240V, fn=50Hz, the voltage at f=30Hz, is

- a. 240V
- b. 144V
- c. 120V
- d. 186V

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AED/INR

Search

10:11 4G 63

Not yet answered

Marked out of 1.00

Flag question

Time left 0:08:48

Induction motor, mechanical characteristics equation is

a. $$T = \frac{3V_{ph}^2 R_2'}{\omega_0 S\left[\left(R_1+R_2'\right)^2+(X_1+X_2')^2\right]}$$

b. $$T = \frac{3V_{ph}^2 R_2'}{\omega_0 S\left[\left(R_1+\frac{R_2'}{S}\right)^2+(X_1+X_2')^2\right]}$$

c. $$T = \frac{3V_{ph}^2 R_2'/s}{\omega_0 S\left[\left(R_1+\frac{R_2'}{S}\right)^2+(X_1+X_2')^2\right]}$$

d. $$T = \frac{3I_2'^2 R_2'}{\omega_0}$$

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DRIVES-1

Time left 0:14:31

Question 1

Not yet answered

Marked out of 1.00

Flag question

Drilling machine torque-speed characteristic is

- a. decreasing non-linear characteristic
- b. independent of speed
- c. decreasing linear characteristic
- d. quadratic characteristic

Clear my choice

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← week 14

10:14 4G 62

Marked out of 1.00

Flag question

Time left 0:06:11

The electric motor heat exchange expression is

a. $p_1 + C\frac{d\theta}{dt} = D\theta$

b. $p_1 + C\frac{d\theta}{dt} = -D\theta$

c. $C\frac{d\theta}{dt} + D\theta = p_1$

d. $C\frac{d\theta}{dt} - D\theta = p_1$

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◄ Quiz 1

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Time left 0:10:22

Question 8

Not yet answered

Marked out of 1.00

Flag question

The hardness of torque-speed characteristic is the relationship between the changes of

a. Speed to current

b. Torque to speed

c. Speed to torque

d. Current to torque

Clear my choice

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10:06 ··· 77%
s2.ebalqa.courses/nfe 24

Not yet answered

Marked out of 1.00

Flag question

Time left 0:09:22

Referring to the multi-quadrant motor operation, the correct statement is

a. the motor can provide motoring and braking operations for forward direction only

b. the motor can provide forward braking when the motor overloaded

c. the motor can provide motoring and braking operations for both forward and reverse directions

d. the motor can provide forward braking when the motor unloaded

Clear my choice

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Finish attempt ...

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Question 7

Not yet answered

Marked out of 1.00

Flag question

Time left 0:10:45

In the electric drive system, the main function of the controller is

a. sensing the feedback variables

b. adapting to the changes of the environment only

c. matching the motor and MTD to meet the load requirements

d. matching the controller and motor to meet the load requirements

Clear my choice

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2025/2024-1-ELE2461-ELECTRICAL DRIVES-1

Time left 0:12:24

Question 4

Not yet answered

Marked out of 1.00

Flag question

The equivalent moment of inertia referred to motor shaft calculation is based on the:

- a. heating (electric) power conservation law
- b. electro-magnetic energy conservation law
- c. mechanical power conservation law
- d. kinetic energy conservation law

Clear my choice

$$= \left(\frac{T_{em1}}{T_{em1}} - \frac{T_{em2}}{T_{em1}}\right) \times 100 = \left[1 - \frac{0.81}{1}\right] \times 100 = 19\%$$

∴ 19% change in starting Torque if voltage is reduced to 90%

Condition for Maximum Torque, Put $\frac{d(T_{em})}{ds} = 0$

$$\therefore \frac{d}{ds}\left[\frac{3 \times 60}{2\pi N_s} \frac{s V^2 R_2}{(R_2^2 + sX_2^2)}\right] = 0 \Rightarrow \boxed{sX_2 = R_2}$$

∴ Condition for Maximum Torque, $\boxed{sX_2 = R_2}$

Since at starting slip, $\boxed{s = 1}$ (Calculated in previous part)

∴ for Maximum Torque at starting, $\boxed{X_2 = R_2}$

∴ Rotor reactance = Rotor resistance → for Maximum starting Torque

Let R_{ex} is resistance added to rotor circuit, therefore rotor resistance, $R_2' = R_2 + R_{ex}$

$$\therefore X_2 = R_2' \Rightarrow R_2 + R_{ex} = X_2$$

$$\Rightarrow 2 + R_{ex} = 5 \Rightarrow R_{ex} = 5 - 2 = 3\Omega$$

$$\therefore \boxed{R_{ex} = 3\Omega}$$

∴ 3Ω resistance should be added to rotor circuit to achieve Maximum torque at starting

Feel free to ask if you have any query. Thank you.

Likes: 0 Dislikes: 0

10:05 4G 67

Time left 0:14:52

Question **1**

Not yet answered

Marked out of 1.00

Flag question

The relationship $K=\sqrt{\frac{1-e^{-|t_{on}/\tau_h + t_{off}/\tau_c|}}{1-e^{-t_{on}/\tau_h}}}$ is used to calculate overloading factor of

a. short time duty cycle

b. continuous duty cycle

c. intermittent periodic duty cycle

d. continuous duty cycle with fluctuating periodic loading

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◀ Quiz 1

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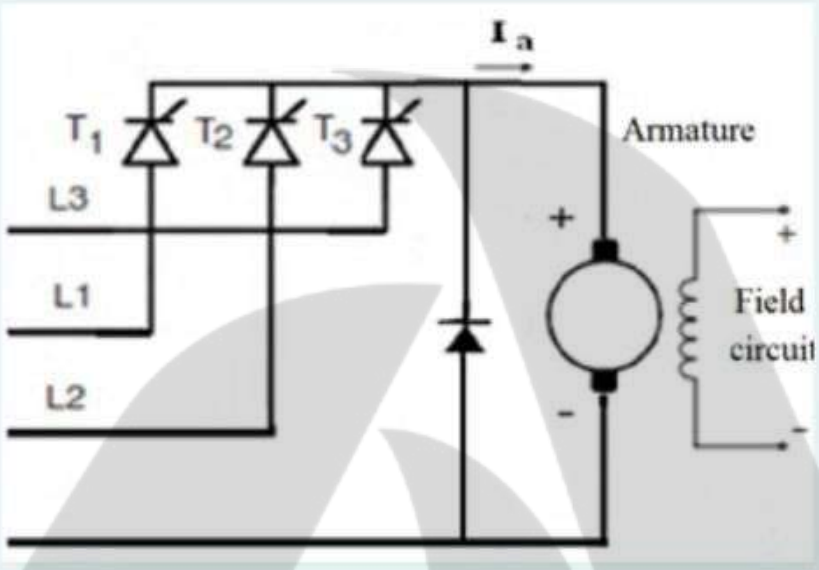
10:16 4G 59

Marked out of 1.00

Time left 0:04:31

Flag question

The half-wave converter drive given in the figure below is



a. a two- quadrant

b. a one- quadrant

c. a three- quadrant

d. a four-quadrant

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Time left 0:10:49

Question **4**

Not yet answered

Marked out of 1.00

Flag question

The temperature rise is related to ambient temperature which value is equal to

a. 20 °C

b. 24°C

c. 30 °C

d. 40 °C

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10:11 4G 63

Not yet answered

Marked out of 1.00

Flag question

Time left 0:08:48

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c. $$T = \frac{3V_{ph}^2 R_2'/s}{\omega_0 S\left[\left(R_1+\frac{R_2'}{S}\right)^2+(X_1+X_2')^2\right]}$$

d. $$T = \frac{3I_2'^2 R_2'}{\omega_0}$$

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10:08 4G 66

Time left 0:12:19

Question **2**

Not yet answered

Marked out of 1.00

Flag question

The average output of a single-phase semiconverter connected to 120V, 50Hz,supply and firing angle of π/2 is

- a. 54.02V
- b. 27V
- c. 0V
- d. 120V

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Next page

◄ Quiz 1

Jump to...

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10:09 4G 64

Question 3

Not yet answered

Marked out of 1.00

Flag question

Time left 0:11:38

With a single –phase full-wave converter in the armature circuit, the average armature voltage is

a. $V_a = \frac{2V_m}{\pi} cos\alpha$

b. $V_a = \frac{V_m}{2\pi} cos\alpha$

c. $\frac{V_m}{\pi}(cos\alpha + 1)$

d. $V_a = \frac{2V_m}{\pi}(cos\alpha + 1)$

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Time left 0:14:52

Question **1**

Not yet answered

Marked out of 1.00

Flag question

The relationship $K=\sqrt{\frac{1-e^{-|t_{on}/\tau_h+t_{off}/\tau_c|}}{1-e^{-t_{on}/\tau_h}}}$ is used to calculate overloading factor of

a. short time duty cycle

b. continuous duty cycle

c. intermittent periodic duty cycle

d. continuous duty cycle with fluctuating periodic loading

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◀ Quiz 1

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10:13 4G 63

Not yet answered

Marked out of 1.00

Flag question

Time left 0:07:21

For single-phase full converter drive of sep.excited dc motor, the reversal of the armature or field allows operation in the

- a. second and third quadrants
- b. first and third quadrant
- c. third and fourth quadrant
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◄ Quiz 1

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Time left 0:12:35

Question **2**

Not yet answered

Marked out of 1.00

Flag question

The active load torque industrial machinery is

- a. compressor
- b. conveyor
- c. hoisting mechanism
- d. turning machine

Clear my choice

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Time left 0:10:45

Question **3**

Not yet answered

Marked out of 1.00

Flag question

Mechanical characteristics of drive system is

a. ω=f(N)

b. ω=f(I)

c. ω=f(t)

d. ω=f(T)

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10:05 78%

2025/2024-1-ELE2461-ELECTRICAL DRIVES-1

Time left 0:09:49

Question 9

Not yet answered

Marked out of 1.00

Flag question

The hardness of series field dc motors torque-speed characteristic is

a. negative

b. zero

c. positive

d. infinity

Clear my choice

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← week 14

10:14 4G 62

Marked out of 1.00

Flag question

Time left 0:06:11

The electric motor heat exchange expression is

a. $p_1 + C\frac{d\theta}{dt} = D\theta$

b. $p_1 + C\frac{d\theta}{dt} = -D\theta$

c. $C\frac{d\theta}{dt} + D\theta = p_1$

d. $C\frac{d\theta}{dt} - D\theta = p_1$

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◄ Quiz 1

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10:04 78%
s2.ebalqa.courses/nfe 24

Question 7

Time left 0:10:45

Not yet answered

Marked out of 1.00

Flag question

In the electric drive system, the main function of the controller is

a. sensing the feedback variables

b. adapting to the changes of the environment only

c. matching the motor and MTD to meet the load requirements

d. matching the controller and motor to meet the load requirements

Clear my choice

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Quiz navigation

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DRIVES-1

Time left 0:14:31

Question 1

Not yet answered

Marked out of 1.00

Flag question

Drilling machine torque-speed characteristic is

- a. decreasing non-linear characteristic
- b. independent of speed
- c. decreasing linear characteristic
- d. quadratic characteristic

Clear my choice

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← week 14

Question 8

Time left 0:04:49

Not yet answered

Marked out of 1.00

Flag question

Motor duty is considered continuous duty (S1), if the motor operates with the following conditions

a. temperature rise during operation under load does not reach steady-state value

b. temperature rise during operation under load reaches steady-state value

c. starting time is not neglected

d. starting losses are not neglected

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◄ Quiz 1

10:06 ··· 77%
s2.ebalqa.courses/nfe 24

Not yet answered

Marked out of 1.00

Flag question

Time left 0:09:22

Referring to the multi-quadrant motor operation, the correct statement is

a. the motor can provide motoring and braking operations for forward direction only

b. the motor can provide forward braking when the motor overloaded

c. the motor can provide motoring and braking operations for both forward and reverse directions

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Clear my choice

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Finish attempt ...

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Quiz navigation

10:16 4G 59

Marked out of 1.00

Flag question

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The temperature limit of insulating materials class c is

- a. 150 °C
- b. > 180 °C
- c. 180 °C
- d. 130 °C

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Finish attempt ...

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10:05 78%
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Time left 0:10:22

Question 8

Not yet answered

Marked out of 1.00

Flag question

The hardness of torque-speed characteristic is the relationship between the changes of

a. Speed to current

b. Torque to speed

c. Speed to torque

d. Current to torque

Clear my choice

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10:03 78%
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2025/2024-1-ELE2461-ELECTRICAL DRIVES-1

Time left 0:12:24

Question 4

Not yet answered

Marked out of 1.00

Flag question

The equivalent moment of inertia referred to motor shaft calculation is based on the:

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c. mechanical power conservation law

d. kinetic energy conservation law

Clear my choice

10:04 78%
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Time left 0:11:16

Question 6

Not yet answered

Marked out of 1.00

Flag question

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a. the third quadrant

b. the fourth quadrant

c. the first quadrant

d. the second quadrant

Clear my choice

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10:02 79%
s2.ebalqa.courses/nfe 24

Time left 0:13:27

Question 2

Not yet answered

Marked out of 1.00

Flag question

Torque equation for rotational motion when inertia is constant

a. $T - T_L = J\frac{d\theta}{dt}$

b. $T - T_L = J\frac{d\omega_m}{dt} + \omega_m\frac{dJ}{dt}$

c. $F_m - F_f = \frac{d(Mv)}{dt}$

d. $T - T_L = J\frac{d^2\theta}{dt^2}$

Clear my choice

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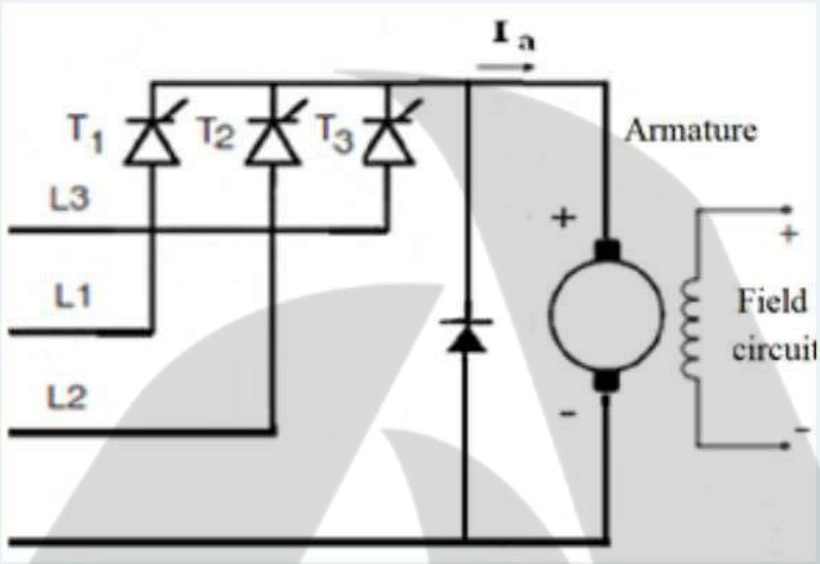
10:16 4G 59

Marked out of 1.00

Time left 0:04:31

Flag question

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a. a two- quadrant

b. a one- quadrant

c. a three- quadrant

d. a four-quadrant

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10:02 78%
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Question 3

Time left 0:12:47

Not yet answered

Marked out of 1.00

Flag question

Torque equation for rotational motion when inertia is variable

a. $F_m - F_f = \frac{d(Mv)}{dt}$

b. $T - T_L = J\frac{d\omega_m}{dt}$

c. $T - T_L = J\frac{d^2\theta}{dt^2}$

d. $T - T_L = J\frac{d\omega_m}{dt} + \omega_m\frac{dJ}{dt}$

Clear my choice

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